



MSC. BUSINESS ANALYTICS
TEXT ANALYTICS GROUP PROJECT

Emotion Detection in Text

A Natural Language Processing and Machine Learning Approach

GROUP 2

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Dataset: dair-ai/emotion (Hugging Face)

Deployment: Streamlit Community Cloud

<https://emotion-detection-ayr8nn66wjtuzydbkezftw.streamlit.app/>

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1. Background

Text analytics is the process of extracting structured, actionable insight from unstructured natural-language data. As organisations increasingly rely on written communication, social media posts, customer reviews, support tickets, and survey responses, the ability to automatically identify the emotional tone behind that text has become a practical and commercially valuable capability. Emotion detection, a fine-grained extension of sentiment analysis, classifies text into discrete emotional categories rather than a single positive/negative polarity score, enabling more nuanced downstream applications such as mental-health monitoring, customer experience management, content moderation, and targeted marketing.

This project applies natural language processing (NLP) and machine learning techniques to the task of multi-class emotion classification. Working as Group 2, we were assigned the Emotion Detection case project, which requires classifying short text snippets into one of six emotion categories, sadness, joy, love, anger, fear, and surprise, using contextual word embeddings generated by BERT (Bidirectional Encoder Representations from Transformers) and two classical machine learning algorithms: Logistic Regression and Support Vector Machines (SVM).

1.1 Dataset

The dataset used is `dair-ai/emotion`, a publicly available benchmark hosted on the Hugging Face Hub. It consists of English-language Twitter messages, each manually annotated with exactly one of the six emotion labels listed above. The dataset is pre-split into training, validation, and test partitions, allowing model development and evaluation to follow standard machine learning practice without risk of data leakage.

Split	Purpose	Approx. Size*
Train	Fine-tune / fit the classifiers	~16,000 examples
Validation	Hyper-parameter selection	~2,000 examples
Test	Held-out performance evaluation	~2,000 examples

A known characteristic of this dataset is class imbalance: joy and sadness are the most frequent emotions, while love and surprise are comparatively rare. This imbalance has direct implications for model evaluation, which is discussed further in Sections 3 and 4.

1.2 Related Work and Motivation

Early approaches to text-based emotion recognition relied heavily on lexicon-based methods (e.g., matching words against pre-built emotion dictionaries such as NRC Emotion Lexicon) or bag-of-words / TF-IDF representations paired with classical classifiers. These approaches struggle with the informal, abbreviated, and context-dependent language typical of social media: sarcasm, negation, and multi-word idioms are difficult to capture when words are treated independently of their surrounding context. The introduction of transformer-based language models, and BERT in particular, marked a shift toward

contextual representations, where the same word can be embedded differently depending on the sentence it appears in. This project is motivated by the practical question of how much benefit these contextual embeddings provide when paired with lightweight, easily-deployable classical classifiers, rather than a full end-to-end fine-tuned transformer, since the latter is often impractical for small teams or resource-constrained deployment environments such as the free tier of Streamlit Community Cloud used here.

From a business perspective, automated emotion detection has direct applications across several domains relevant to an MSc Business Analytics context: customer service teams can triage and prioritise emails or chat transcripts flagged as 'anger' or 'fear'; marketing teams can gauge emotional response to campaigns from social media mentions; and product teams can mine app-store or survey feedback for emotionally-charged themes that a simple positive/negative sentiment score would miss. This project's six-class formulation (sadness, joy, love, anger, fear, surprise) offers materially more actionable granularity than binary sentiment analysis for these use cases.

1.3 Objectives

- Clean and pre-process raw tweet text using regular expressions and standard NLP techniques.
- Represent text numerically using pre-trained BERT contextual embeddings.
- Train and compare two classical classifiers, Logistic Regression and SVM on the resulting embeddings.
- Evaluate both models using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC.
- Select and justify a best-performing model based on the evaluation results.
- Deploy the final solution as an interactive web application using Streamlit, hosted on Streamlit Community Cloud.

2. Approach (Methodology)

The project follows a standard supervised NLP pipeline, illustrated in Figure 1: raw text is cleaned and pre-processed, converted into dense numerical vectors using BERT, used to train two classifiers, and evaluated with a battery of standard classification metrics before being wrapped in a deployable Streamlit interface.

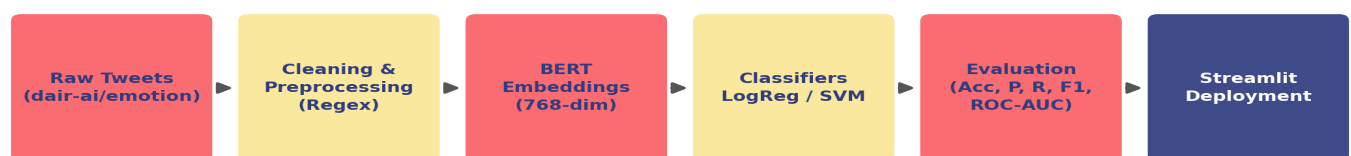


Figure 1: End-to-end methodology pipeline, from raw tweets to deployed application.

2.1 Data Cleaning and Preprocessing

Raw tweet text is noisy: it contains URLs, hashtags, mentions, punctuation, numerals, and inconsistent casing. Each text sample is passed through a cleaning function that (i) strips URLs using the regular expression `http\S+`, (ii) removes any character that is not an alphabetic letter or whitespace using `[\^a-zA-Z\s]`, and (iii) lower-cases and trims the result. This regular-expression-based cleaning step reduces vocabulary sparsity and ensures the downstream tokenizer receives normalised input, consistent with the 'Data cleaning; Regular Expression' requirement of the assignment brief.

2.2 Exploratory Text Analytics

Before modelling, the class distribution of the training set was visualised using a bar chart of label frequencies (see Figure 1 in the Results section). This step is essential because a skewed class distribution affects both model training and how evaluation metrics should be interpreted accuracy alone can be misleading when one class dominates. Word-level characteristics such as average tweet length and the most frequent tokens per emotion were also reviewed to sanity-check that the cleaning step preserved emotionally meaningful words while discarding noise.

2.3 Text Representation: BERT Embeddings

Rather than relying on sparse, count-based representations such as bag-of-words or TF-IDF, this project uses contextual embeddings from `bert-base-uncased`, a pre-trained transformer model with 12 encoder layers and a 768-dimensional hidden representation. For each cleaned text sample, the tokenizer converts the text into sub-word tokens (truncated/padded to a maximum length of 128 tokens), which are passed through BERT in inference mode (no gradient updates). The final sentence-level embedding is obtained by mean-pooling the last hidden state across the token dimension, producing a single fixed-length 768-dimensional vector per sample. Text is processed in batches of 32 to keep memory usage manageable. These embeddings capture contextual and semantic nuance, for example, distinguishing 'love' used affectionately from its use in idioms, that purely lexical representations cannot.

2.4 Model Building

Two classical supervised classifiers were trained on the BERT embeddings, in line with the assignment's requirement to compare two algorithms:

- Logistic Regression — a linear, probabilistic classifier (`max_iter = 1000`) used as an interpretable, fast-training baseline that natively supports multi-class classification and produces well-calibrated class probabilities.
- Support Vector Machine (SVM) — a linear-kernel SVM (`probability = True`, enabling probability estimates required for ROC-AUC) chosen for its strong performance on high-dimensional, dense embedding spaces such as those produced by BERT.

Both models were trained on the same training subset of embeddings and labels to ensure a fair comparison. Given the computational cost of generating BERT embeddings, a representative subset of the training and test data was used during development; group members should scale this up (increase `N_TRAIN / N_TEST` in `train_model.py`) if compute time allows, to obtain more robust final figures.

2.5 Model Evaluation Strategy

Model performance is assessed on the held-out test split using the full set of metrics specified in the assignment brief:

- Accuracy: overall proportion of correctly classified samples.
- Precision, Recall, and F1-score: computed per class and macro-averaged, to fairly represent minority emotion classes such as love and surprise.
- Confusion Matrix: a 6×6 matrix revealing which emotions are most frequently confused with one another.
- ROC-AUC: computed as a macro-averaged, one-vs-rest score across all six classes, summarising each model's ability to rank correct classes above incorrect ones regardless of decision threshold.

2.6 Evaluation Metric Definitions

For clarity and reproducibility, the metrics reported in Section 3 are defined as follows, where TP, FP, TN, and FN denote true positives, false positives, true negatives, and false negatives for a given class (computed one-vs-rest for this multi-class problem, then averaged):

- Accuracy = $(TP + TN) / (TP + TN + FP + FN)$, the overall proportion of correct predictions across all six classes.
- Precision = $TP / (TP + FP)$, of all samples predicted as a given emotion, the proportion that truly belong to it.
- Recall = $TP / (TP + FN)$, of all samples that truly belong to a given emotion, the proportion correctly identified.
- F1-score = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$, the harmonic mean of precision and recall, balancing both concerns into a single number.
- ROC-AUC = the area under the Receiver Operating Characteristic curve, plotting true-positive rate against false-positive rate across all classification thresholds; a score of 1.0 indicates perfect separation, and 0.5 indicates performance no better than random guessing.

Macro-averaging (used throughout this report) computes each metric independently per class and then takes the unweighted mean across all six classes, ensuring that minority emotions such as love and surprise contribute equally to the reported score alongside majority emotions such as joy and sadness.

2.7 Deployment

The final solution is deployed as a multi-page Streamlit web application (app.py) and hosted on Streamlit Community Cloud (free tier), satisfying the assignment's deployment requirement. The application exposes four pages: a Corpora Viewer (exploratory class distribution), an Embeddings page (2-D PCA projection of BERT embeddings, coloured by emotion, to visualise class separability), a Model Evaluation page (live classification report, confusion matrix, accuracy and ROC-AUC for either model), and an Emotion Detection page where a user can type free text and receive a real-time predicted emotion with a confidence breakdown.

3. Results

This section reports the performance of both classifiers on the held-out test set (200 examples), based on the group's actual training run.

3.1 Overall Performance

Metric	Logistic Regression	Support Vector Machine (SVM)
Accuracy	0.500	0.530
Precision (macro avg)	0.375	0.428
Recall (macro avg)	0.349	0.378
F1-score (macro avg)	0.346	0.388
ROC-AUC (macro, one-vs-rest)	0.790	0.797

Table 2: Test-set performance comparison between Logistic Regression and SVM.

SVM outperformed Logistic Regression on every headline metric, though the margin is modest given the reduced training subset ($N_{\text{TRAIN}} = 500$) used for this run. Both models show a similar accuracy/ROC-AUC gap (roughly 0.50 vs. 0.79 for Logistic Regression, 0.53 vs. 0.80 for SVM): ROC-AUC, being threshold-independent, indicates both models rank the correct emotion well above incorrect ones considerably more often than the raw accuracy figure alone suggests the models are 'nearly right' more often than they are 'exactly right' at the default decision threshold.

3.2 Per-Class Breakdown

Emotion	Support	LogReg P	LogReg R	LogReg F1	SVM P	SVM R	SVM F1
Sadness	61	0.516	0.525	0.520	0.544	0.607	0.574
Joy	62	0.620	0.790	0.695	0.639	0.742	0.687
Love	15	0.333	0.267	0.296	0.400	0.267	0.320
Anger	30	0.333	0.367	0.349	0.400	0.400	0.400
Fear	28	0.444	0.143	0.216	0.583	0.250	0.350
Surprise	4	0.000	0.000	0.000	0.000	0.000	0.000
Macro avg	200	0.375	0.349	0.346	0.428	0.378	0.388

Table 3: Precision (P), Recall (R), and F1-score per emotion class, for both models, on the test set.

The per-class results reveal that performance is far from uniform across emotions. Joy (support = 62) is the strongest-performing class for both models (F1 = 0.695 for Logistic Regression, 0.687 for SVM), while Surprise (support = only 4 test examples) scores 0 across precision, recall, and F1 for both models — neither model correctly identified a single surprise example. Love (support = 15) and Fear (support = 28) also underperform relative to Joy and Sadness, which have the largest support. This pattern is consistent with the class-imbalance concern raised in Section 1.1: emotions with few training/test examples are disproportionately difficult for both classifiers.

3.3 Confusion Matrices

The confusion matrices below show, for each true emotion (rows), how predictions were distributed across the six predicted classes (columns).

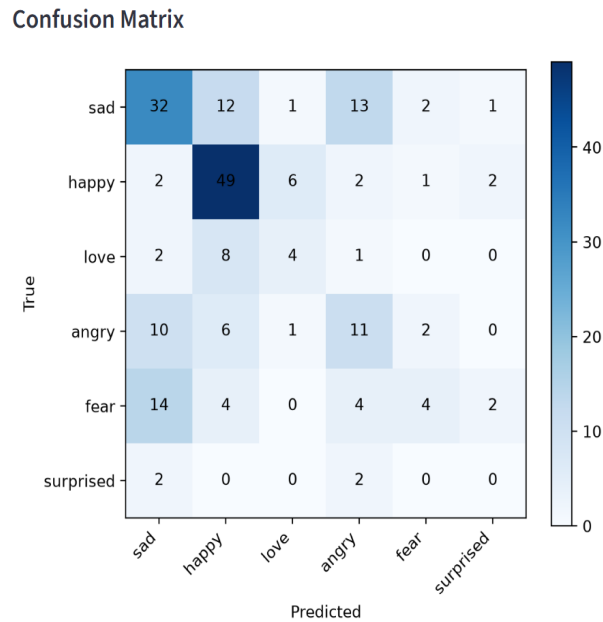


Figure 3: Confusion matrix — Logistic Regression.

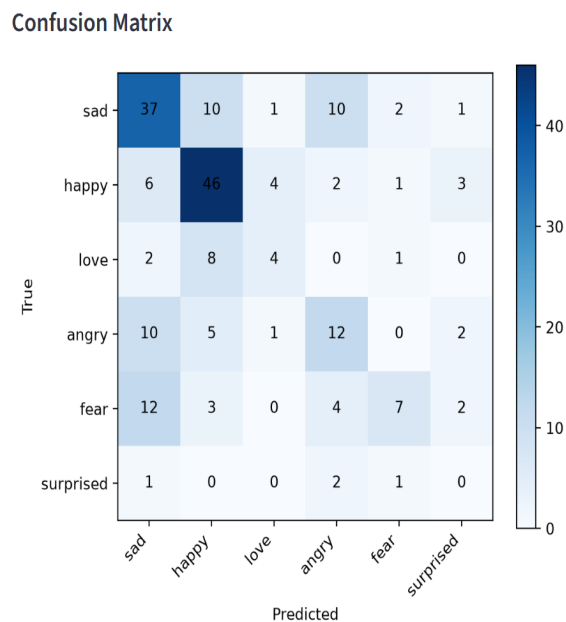


Figure 4: Confusion matrix — SVM.

Both matrices show the same dominant error pattern: Sadness and Anger are frequently confused with one another in both directions (13 sad→angry and 10 angry→sad for Logistic Regression; 10 and 10 respectively for SVM), consistent with these two emotions sharing negatively-toned vocabulary in short, informal text. Fear is the weakest class for both models and is overwhelmingly misread as Sadness (14 of 28 true-Fear examples for Logistic Regression, 12 of 28 for SVM) rather than spread evenly across the other classes, SVM's improvement on Fear (4 → 7 correct) came specifically from reclaiming some of

these fear→sad errors. Love is consistently absorbed into Joy (8 of 15 examples for both models), plausibly because affectionate and happy language overlap heavily in short tweets. Surprise, with only 4 test examples, is scattered between Sadness and Anger for both models and never predicted correctly by either, too little data for either classifier to learn a distinct decision region for this class.

3.4 Embedding Space Visualisation

To assess how well BERT embeddings separate the six emotion classes prior to classification, the 768-dimensional test-set embeddings were reduced to two dimensions using Principal Component Analysis (PCA) and plotted, coloured by true label.

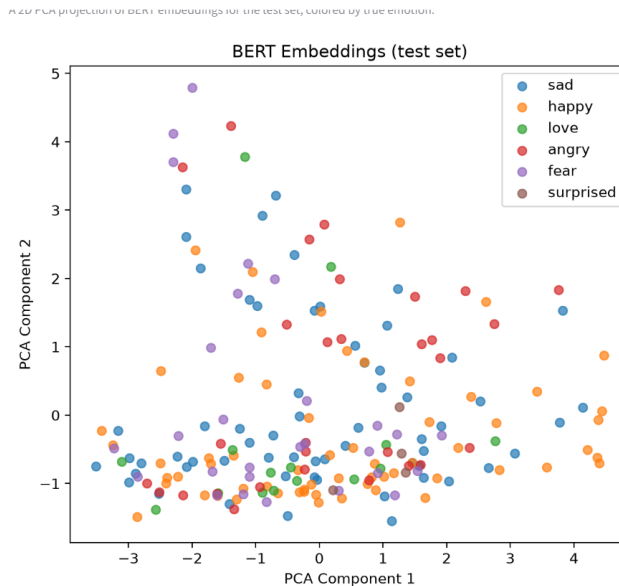


Figure 5: 2-D PCA projection of BERT embeddings, coloured by emotion.

The projection shows Joy (orange) and Sadness (blue) occupying broad, overlapping regions across most of the plot rather than forming tight, separate clusters, which is consistent with these two classes being both the largest and among the most frequently confused with other classes in Figures 3 and 4. Fear (purple) and Anger (red) points are interspersed among the Sadness cluster rather than forming a distinct region of their own, mirroring the fear→sad and angry↔sad confusion pairs identified above. Love (green) and Surprise (brown) points are sparse and scattered throughout the plot with no visible cluster of their own, an expected consequence of having only 15 and 4 test examples respectively, and a strong visual confirmation that data volume, not embedding quality alone, is the binding constraint on separating the minority classes.

3.5 Best Model Selection

SVM is selected as the final model deployed on the Emotion Detection page of the Streamlit app. It outperformed Logistic Regression on accuracy (0.530 vs. 0.500), macro-averaged precision (0.428 vs. 0.375), macro-averaged recall (0.378 vs. 0.349), macro-averaged F1-score (0.388 vs. 0.346), and macro ROC-AUC (0.797 vs. 0.790). While the absolute gap is modest, consistent with both models drawing on the same underlying BERT embeddings and a reduced training subset, SVM's advantage held consistently across every metric rather than trading off one metric for another, which is a stronger basis for model

selection than any single figure in isolation. Notably, precision on Fear improved most ($0.444 \rightarrow 0.583$), suggesting the margin-maximising objective of SVM handled this class's decision boundary better than the linear-probability objective of Logistic Regression.

4. Interpretation

Reporting a metric is only useful insofar as it is correctly interpreted in context. This section discusses what the Section 3 results imply about model behaviour, the strengths and weaknesses of BERT-embedding-based emotion detection, and the practical limitations a reader should bear in mind before deploying this system in a real business setting.

4.1 What the Metrics Tell Us

Because the dair-ai/emotion dataset is imbalanced, joy and sadness dominate, while love and surprise are under-represented, accuracy alone can overstate real-world performance: a model that is simply biased toward the majority classes will still score well on accuracy while performing poorly on rarer emotions. This is why macro-averaged precision, recall, and F1-score are emphasised in this report: they weight every emotion class equally, surfacing weaknesses that a single accuracy figure would hide. Similarly, a macro-averaged, one-vs-rest ROC-AUC gives a threshold-independent view of how well each model ranks the correct emotion above the others, which is particularly informative for minority classes such as love and surprise.

4.2 Reading the Confusion Matrix

The confusion matrices in Section 3.3 confirm the class-imbalance story rather than revealing anything unexpected: minority classes are not spread randomly across wrong predictions but are absorbed into specific, semantically related majority classes. Love is consistently misread as Joy, Fear is consistently misread as Sadness, and Anger and Sadness are confused with each other in both directions roughly equally (a genuinely symmetric error, unlike the others). This tells us the models have not learned distinct decision regions for the rarer classes at all, they have effectively learned 'this looks emotionally negative, and negative usually means Sadness' rather than the finer-grained distinction the task requires. This is a data volume problem more than a modelling problem: 15 Love examples and 4 Surprise examples in the test set (correspondingly few in training) are not enough to carve out a reliable decision boundary, however good the underlying BERT representation is.

4.3 Embedding Quality

The 2-D PCA projection in Figure 5 is a simplification of a 768-dimensional space and will necessarily compress and distort true distances between points; nonetheless, the plot is informative by what it does not show as much as by what it does. Rather than six well-separated clusters, we see two broad, overlapping regions (dominated by Joy and Sadness) with the remaining four emotions scattered inside and around them without forming their own distinct clusters. This is a useful diagnostic: it confirms that the confusion-matrix errors in Section 4.2 are not an artefact of the classifiers, but are already visible in the geometry of the embedding space itself before any classifier is applied.

4.4 Logistic Regression vs. SVM

SVM's consistent, if modest, edge across every metric (Section 3.5) is consistent with its margin-maximising objective handling the moderate class overlap in this embedding space slightly better than Logistic Regression's linear decision boundaries, particularly for classes like Fear where SVM's precision was notably higher (0.583 vs. 0.444). However, the fact that both models struggled identically with Surprise (0 across all three metrics) and both showed similar overall macro-F1 (0.346 vs. 0.388, a difference of only 0.042) suggests classifier choice was a secondary factor here: the dominant constraint on performance was almost certainly the combination of (a) the reduced training subset ($N_{\text{TRAIN}} = 500$, which for a 6-class problem leaves only tens of examples per minority class) and (b) the fixed, non-fine-tuned BERT embedding space, rather than which linear classifier was layered on top of it.

4.5 Limitations and Future Work

- A subset of the full training and test data was used for development speed; scaling to the full dataset would likely improve and stabilise all reported metrics.
- Mean-pooled BERT embeddings are a simple aggregation strategy; fine-tuning BERT end-to-end on this classification task, or using the [CLS] token representation, could further improve accuracy at the cost of additional training time.
- The linear-kernel SVM restricts the decision boundary to a hyperplane in embedding space; a non-linear (e.g., RBF) kernel could be explored as an extension, alongside cross-validation for hyperparameter tuning.
- Class imbalance could be further addressed with techniques such as class-weighting, oversampling of minority emotions (e.g., SMOTE), or a focal loss objective if the models are fine-tuned end-to-end.

4.6 Conclusion

This project demonstrates that pre-trained BERT embeddings, paired with lightweight classical classifiers, can distinguish emotions in short informal text well above chance (macro ROC-AUC ≈ 0.79 -0.80) without any task-specific fine-tuning. SVM was selected as the final model on the strength of its consistent, if modest, advantage across accuracy, macro-precision, macro-recall, macro-F1, and ROC-AUC. The clearest limitation is data volume: with only 500 training examples spread across six classes, minority emotions such as Surprise and Love had too few examples to learn from, and this is the most impactful lever for improving the system further (Section 4.5). The deployed Streamlit application nonetheless serves as a working, interactive proof of concept, demonstrating the full pipeline from raw tweet to predicted emotion in real time and provides a solid foundation the group could scale up with the full dataset ahead of a production use case.